

AnalystLab Africa Internship Programme
FinTrust Digital Bank – Week 1
FinTrust Business & Data Intelligence Assessment

Track: Data Analytics
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Week: Week 1 – Understand & Plan

BUSINESS UNDERSTANDING

1.1 Business Problem

FinTrust Digital Bank wants to make better use of its customer and transaction data to understand customer behaviour, transaction activity, operational performance and risk-related patterns. The project also aims to support better decision-making and improve digital customer support.

For the Data Analytics track, my role is to examine the available customer and transaction data and identify useful patterns and insights that can be presented in a clear way for business decision-making.

1.2 Why the Problem Is Important

FinTrust has a large amount of customer and transaction information, but having data alone does not automatically provide useful business insights. The data needs to be analysed to understand what is happening across customers, transactions and banking channels.

This analysis can help FinTrust understand how customers use its services, which transaction channels are being used, how transactions are performing, and where patterns that may require further attention exist. These insights can provide a stronger basis for operational and customer-focused decisions.

1.3 Contribution of Data Analytics

Data Analytics can help FinTrust turn its raw customer and transaction records into information that is easier to understand and use.

For this project, I will use the available data to explore customer characteristics and behaviour, transaction activity and value, channel usage, transaction outcomes and risk-review patterns.

The findings can then be summarized using KPIs and visualisations so that important patterns can be identified more easily.

1.4 Business Questions

Based on the available data and the purpose of the project, the analysis should help FinTrust answer questions such as:

1. What are the main characteristics and segments of FinTrust's customers?
2. How does transaction activity change over time?
3. Which transaction types generate the highest transaction volumes and values?
4. Which transaction channels are used most frequently?
5. What proportion of transactions are successful, failed, reversed or pending?
6. How does transaction behaviour differ across customer segments?

7. What patterns can be observed among transactions flagged for risk review?
8. Are there differences in transaction activity or outcomes across channels and locations?

These questions stay within the areas AnalystLab specifically expects the Data Analytics track to investigate: customer behaviour, transaction activity and value, channels, transaction status and risk-related patterns.

1.5 Decisions the Analysis Could Support

The analysis could support decisions relating to customer engagement, transaction operations, digital banking channels, service improvement and risk monitoring.

For example, management could use the analysis to understand which channels customers use most, identify transaction types or channels associated with unsuccessful transactions, compare transaction behaviour across customer segments and identify patterns in transactions that have been flagged for risk review.

At this stage, these are areas the analysis could support rather than conclusions from the data. The actual findings will come from the analysis performed in later stages of the project.

1.6 Potential Stakeholders

The analytical outputs could be useful to different stakeholders within FinTrust. Senior management could use high-level KPIs to monitor business activity, while operations teams could use transaction information to understand transaction performance and operational issues.

Customer service or customer experience teams could benefit from information about customer behaviour and preferred channels. Risk-related teams could use the analysis to examine transactions flagged for review, while digital banking or product teams could use channel and engagement information to better understand how customers interact with FinTrust's digital services.

1.7 Expected Data Analytics Output

The Data Analytics track will produce business insights supported by KPIs and visualisations. The eventual output is expected to make FinTrust's customer and transaction information easier for stakeholders to understand and use in decision-making.

For Week 1, my focus is on understanding the business problem and available data, identifying useful analytical questions and KPIs, and planning the structure of the analysis and dashboard.

REVIEW OF RELEVANT RESOURCES

2.1 FinTrust Customer Data

The `FinTrust_Customer_Data.csv` file contains customer-level information such as Customer ID, customer name, age, gender, city, customer segment, account type, tenure, digital engagement score, monthly income band, preferred channel and account status.

I will use this dataset to understand the characteristics of FinTrust's customers and explore areas such as customer segments, account types, digital engagement and channel preferences. It will also be linked to the transaction data using `Customer_ID` so that customer characteristics can be compared with transaction behaviour.

Limitation: The dataset is synthetic and was created for educational purposes, so the findings from this project should not be treated as representing real FinTrust customers or actual banking activity. AnalystLab states that the supplied customer, transaction and financial information is synthetic.

2.2 FinTrust Transaction Data

The `FinTrust_Transaction_Data.csv` file contains transaction-level information, including Transaction ID, Customer ID, transaction date and time, transaction type, amount in NGN, channel, device type, location, international transaction status, transaction status and risk review flag.

I will use this dataset to examine transaction activity and value, transaction types, channel usage, transaction outcomes and risk-related patterns. It can also be combined with the customer dataset to understand how transaction behaviour differs across customer groups.

Limitation: During my initial data profiling, I identified missing values in `Device_Type` and `Location`. There are **96 missing values in each field, representing 0.8% of the 12,000 transaction records**. These values have not been cleaned or replaced at this stage because Week 1 focuses on understanding the data rather than extensive cleaning.

2.3 FinTrust Data Dictionary

The `FinTrust_Data_Dictionary.csv` provides descriptions and expected data types for the fields contained in the customer and transaction datasets. It also provides useful information about how individual variables should be interpreted.

I used the data dictionary as a reference while profiling the datasets, particularly when identifying variable types and understanding the relationship between the two datasets. For example, it identifies `Customer_ID` as the key that connects customer records with transaction records.

Limitation: The data dictionary explains what the variables are intended to represent, but it does not provide analytical findings. The actual datasets still need to be examined to confirm their structure and identify data-quality issues.

2.4 Relationship Between Resources

The Customer and Transaction datasets are related through the `Customer_ID` field. This creates a **one-to-many relationship**, where one customer can be associated with multiple transactions.

I verified this relationship during profiling. The Customer dataset contains **1,500 unique customers**, and all **1,500 customers appear in the Transaction dataset**. I also found **zero transaction Customer IDs without a corresponding record in the Customer dataset**.

This relationship will allow customer information to be combined with transaction information during later analysis. For example, it will make it possible to investigate whether transaction behaviour differs between customer segments.

DATA UNDERSTANDING AND PROFILING

I used **Python and Pandas** to carry out an initial review of the FinTrust Customer and Transaction datasets. The purpose of this stage was to understand the structure and quality of the data before carrying out detailed analysis.

I checked the dataset dimensions, field names, data types, missing values, duplicates, variable classifications and the relationship between the two datasets. I also reviewed categorical values and numerical ranges for obvious quality issues. No extensive data cleaning was performed at this stage, in line with the Week 1 instructions.

3.1 Dataset Dimensions

The first step was to determine the size of each dataset.

DATASET	RECORDS	COLUMNS
Fintrust Customers data	1500	12
Fintrust Transaction data	12000	11

The Customer dataset contains **1,500 customer records**, while the Transaction dataset contains **12,000 transaction records**.

3.2 Customer Dataset Structure

The Customer dataset contains 12 fields:

Field	Data Type	Class
Customer_ID	String	Identifier
Customer_Name	String	Descriptive
Age	Integer	Numerical
Gender	String	Categorical
City	String	Categorical
Customer_Segment	String	Categorical
Account_Type	String	Categorical
Tenure_Months	Integer	Numerical
DigitalEngagement_Score	Float	Numerical
Monthly_Income_Band	String	Categorical
Preferred_Channel	String	Categorical
Account_Status	String	Categorical

The main numerical variables are **Age**, **Tenure_Months** and **Digital_Engagement_Score**. The remaining analytical fields are mainly categorical, while **Customer_ID** serves as the unique customer identifier.

3.3 Transaction Dataset Structure

The Transaction dataset contains 11 fields:

Field	Data Type	Classification
Transaction_ID	String	Identifier
Customer_ID	String	Identifier / Join Key
Transaction_DateTime	String	Date/Time*
Transaction_Type	String	Categorical
Amount_NGN	Float	Numerical

Channel	String	Categorical
Device_Type	String	Categorical
Location	String	Categorical
International_Transaction	String	Categorical
Transaction_Status	String	Categorical
Risk_Review_Flag	String	Categorical

Amount_NGN is the main numerical variable in the Transaction dataset.

One issue identified during profiling is that **Transaction_DateTime** was loaded by Pandas as a **string**, even though it represents date and time information. This will need to be converted to a datetime data type before carrying out time-based analysis.

3.4 Missing-Value Assessment

I checked both datasets for missing values.

The Customer dataset contained **no missing values** across its 1,500 records.

In the Transaction dataset, missing values were found in two fields:

Field	Missing Values	Percentage
Device_Type	96	0.8%
Location	96	0.8%
All other fields	0	0%

The missing values have not been removed or replaced at this stage. They will be considered during the data-preparation stage when the appropriate treatment can be determined.

3.5 Duplicate Assessment

I also checked for duplicate records and duplicate primary identifiers.

Check	Result
Duplicate Customer rows	0
Duplicate Transaction rows	0
Duplicate Customer_IDs	0
Duplicate Transaction_IDs	0

No duplicate rows were identified in either dataset. **Customer_ID** is unique within the Customer dataset, while **Transaction_ID** is unique within the Transaction dataset.

3.6 Dataset Relationship

The two datasets are connected through **Customer_ID**. My relationship check produced the following results:

Check	Result
Unique customers in Customer dataset	1,500
Unique customers appearing in Transaction dataset	1,500
Transaction Customer_IDs without a matching customer	0

This confirms a **one-to-many relationship** between the datasets: one customer can have multiple transactions.

It also shows that all customers represented in the Transaction dataset have corresponding records in the Customer dataset, so no unmatched customer IDs were identified.

3.7 Descriptive Statistical Analysis

I performed descriptive statistical analysis on the numerical variables to better understand their distributions and identify potential outliers. I compared the mean and median and used the Interquartile Range (IQR) method to identify values falling outside the usual $1.5 \times \text{IQR}$ boundaries.

Variable	Mean	Median	Potential Outliers	Outlier %
Age	41.59	42.00	0	0.00%
Tenure (Months)	49.86	51.00	0	0.00%
Digital Engagement Score	67.95	68.00	4	0.27%
Transaction Amount (NGN)	₦ 46,706.45	₦ 10,305.94	1,474	12.28%

The mean and median values for Age, Tenure and Digital Engagement Score are relatively close. This suggests that either measure provides a reasonable overview of the centre of these variables, although the median remains less sensitive to extreme values.

Transaction Amount shows a different pattern. The mean transaction amount is **~~₦~~46,706.45**, compared with a much lower median of **~~₦~~10,305.94**. The IQR assessment also identified **1,474 transactions (12.28%)** as potential outliers. This suggests that the transaction amount distribution is right-skewed and influenced by relatively high-value transactions.

For this reason, the **median transaction amount is more representative of a typical transaction value than the mean**. However, transactions identified by the IQR method will not automatically be treated as errors or removed. High-value transactions may be valid banking activity and should be investigated further before deciding how they should be handled.

3.8 Preliminary Categorical Assessment

I reviewed the unique values and frequencies of the categorical variables to check for obvious inconsistencies.

For example, the Customer dataset contains four customer segments: **Everyday, Premium, Student and SME**. Preferred channels are **Mobile App, Web and USSD**, while account statuses are **Active, Dormant and Restricted**.

The Transaction dataset includes transaction statuses of **Successful, Failed, Reversed and Pending**, while the available channels include **Mobile App, POS, Web, ATM and USSD**.

No obvious spelling or capitalization inconsistencies were identified in the categorical values reviewed.

3.9 Identified Data-Quality Issues

Based on the initial profiling, three main issues require attention before detailed analysis: No extensive corrections were made during Week 1 because AnalystLab specifically asks for obvious data-quality issues to be identified without extensive cleaning.

Data-Quality Issue	Finding	Planned Treatment
Missing Device Type	96 values (0.8%)	Assess appropriate treatment during data preparation
Missing Location	96 values (0.8%)	Assess appropriate treatment during data preparation
Transaction_DateTime data type	Imported as string	Convert to datetime before time-based analysis
Potential high-value transaction outliers	1,474 records (12.28%) identified using IQR	Investigate before deciding whether any treatment is necessary

3.10 Data Understanding Summary

The initial profiling shows that the FinTrust datasets are well structured enough to proceed with further analysis.

The Customer dataset contains **1,500 records and 12 fields**, while the Transaction dataset contains **12,000 records and 11 fields**. No duplicate records or duplicate primary identifiers were found, and all customer IDs appearing in the Transaction dataset have matching records in the Customer dataset.

The main issues identified were **0.8% missing values in both Device_Type and Location**, as well as the need to convert **Transaction_DateTime** from string to datetime before time-based analysis.

These findings provide a clear starting point for the next stages of the FinTrust analysis.

4. ANALYTICAL QUESTIONS

Following the initial review and profiling of the FinTrust datasets, I developed the following analytical questions to guide the next stage of the project. The questions focus on areas that can be investigated using the available customer and transaction data.

1. **How is FinTrust's customer base distributed across customer segments, account types and account statuses?**

This will provide an overview of the customer base and help identify the major customer

groups using FinTrust's services.

2. How does transaction volume and value change over time?

This will help identify patterns in transaction activity and determine whether there are noticeable periods of higher or lower activity.

3. Which transaction types contribute the most to total transaction volume and transaction value?

This will show how activities such as transfers, card purchases, bill payments, withdrawals, deposits and airtime/data transactions contribute to overall transaction activity.

4. Which transaction channels are most frequently used, and how does transaction value vary across these channels?

This will help compare the use of Mobile App, POS, Web, ATM and USSD and determine how transaction activity differs between channels.

5. What is the distribution of successful, failed, reversed and pending transactions, and does transaction status vary across channels or transaction types?

This will provide insight into transaction outcomes and help identify areas where unsuccessful transactions may require further investigation.

6. How does transaction behaviour differ across customer segments?

This will combine the Customer and Transaction datasets to examine whether segments such as Everyday, Premium, Student and SME customers show differences in transaction frequency or value.

7. What characteristics are associated with transactions flagged for risk review?

This will examine factors available in the dataset—such as transaction amount, transaction type, channel, international transaction status and location—to identify patterns among transactions where `Risk_Review_Flag = Yes`.

8. How does customers' digital engagement relate to their transaction behaviour and preferred transaction channels?

This will explore whether differences in `Digital_Engagement_Score` are associated with transaction activity and channel usage.

5. KPI PLANNING

Based on the analytical questions developed for the project, I identified the following KPIs to provide a clear overview of FinTrust's customer and transaction activity. These KPIs will guide the detailed analysis and dashboard development in the later stages of the project.

KPI	Definition	Why It Matters	Required Data
Total Customers	Number of unique customers in the customer dataset.	Provides an overview of the size of FinTrust's customer base and can support customer-segment analysis.	Customer_ID
Total Transactions	Total number of transaction records.	Shows the overall level of transaction activity processed through FinTrust.	Transaction_ID
Total Transaction Value	Sum of the value of all transactions in NGN.	Shows the overall monetary value of transactions recorded during the analysis period.	Amount_NGN
Median Transaction Value	Middle transaction amount when all transaction values are arranged from lowest to highest.	Provides a better indication of a typical transaction value because preliminary profiling showed that transaction amounts are right-skewed and influenced by high-value transactions.	Amount_NGN
Successful Transaction Rate	Percentage of all transactions with a status of Successful . Formula: $\text{Successful Transactions} \div \text{Total Transactions} \times 100$.	Helps monitor transaction outcomes and provides a high-level indication of how frequently recorded transactions are completed successfully.	Transaction_ID, Transaction_Status

Failed Transaction Rate	Percentage of all transactions with a status of Failed . Formula: $\text{Failed Transactions} \div \text{Total Transactions} \times 100$.	Helps identify the proportion of transactions that fail and provides a basis for investigating differences across channels or transaction types.	Transaction_ID, Transaction_Status
Most Used Transaction Channel	Transaction channel with the highest number of transactions.	Helps identify the channel through which customers most frequently conduct transactions and supports channel-usage analysis.	Transaction_ID, Channel
Risk Review Rate	Percentage of transactions where Risk_Review_Flag = Yes . Formula: $\text{Risk-Reviewed Transactions} \div \text{Total Transactions} \times 100$.	Provides a high-level measure of how frequently transactions are flagged for risk review and supports further investigation of risk-related patterns.	Transaction_ID, Risk_Review_Flag

6. DASHBOARD PLANNING / WIREFRAME

The proposed FinTrust dashboard will provide a clear overview of customer and transaction activity while allowing users to explore transaction performance, customer behaviour, channel usage and risk-review patterns. The dashboard will be designed so that high-level KPIs are visible first, followed by more detailed visualisations.

6.1 Proposed Dashboard Sections

I plan to organise the dashboard into four main sections:

Overview: A high-level summary of FinTrust's customer base, transaction activity and transaction value.

Transaction Performance: Analysis of transaction trends, transaction types and transaction outcomes.

Customer & Channel Analysis: Customer segments, digital engagement and how customers use different transaction channels.

Risk Review Analysis: Patterns among transactions flagged for risk review, including transaction amounts, channels, types and international transaction status.

6.2 Proposed KPI Cards

The top section of the dashboard will contain KPI cards showing the most important summary measures:

- Total Customers
- Total Transactions
- Total Transaction Value
- Median Transaction Value
- Successful Transaction Rate
- Failed Transaction Rate
- Most Used Transaction Channel
- Risk Review Rate

6.3 Proposed Charts and Visualisations

Visual	Purpose
Line Chart – Transaction Activity Over Time	Show how transaction volume and/or value changes over time.
Bar Chart – Transactions by Type	Compare transaction volume and value across transfers, card purchases, bill payments, withdrawals, deposits and airtime/data transactions.
Bar/Column Chart – Transactions by Channel	Compare the use of Mobile App, POS, Web, ATM and USSD.
Donut/Bar Chart – Transaction Status	Show the proportion of successful, failed, reversed and pending transactions.

Bar Chart – Customer Segments

Compare the size and transaction behaviour of Everyday, Premium, Student and SME customers.

Bar Chart – Risk Review by Transaction Type/Channel

Compare risk-review patterns across transaction types or channels.

Chart – Digital Engagement and Transaction Behaviour

Explore how digital engagement relates to customer transaction activity.

6.4 Proposed Filters

The dashboard should allow users to narrow the analysis using relevant filters such as:

Date/Time Period, Customer Segment, Transaction Type, Channel, Transaction Status, Location, International Transaction, and Risk Review Flag.

These filters will allow users to move from a general overview to more specific questions—for example, examining failed Mobile App transactions or risk-reviewed international transactions.

6.5 Proposed Dashboard Layout

For the actual **wireframe**, you can put this into your Google Doc using a simple table:

FINTRUST ANALYTICS DASHBOARD			
Filters: Date Segment Channel Type Status Risk			
TOTAL CUSTOMERS	TOTAL TRANSACTIONS	TOTAL TRNX VALUE	MEDIAN TRNX VALUE
SUCCESS RATE	FAILED RATE	MOST USED CHANNEL	RISK REVIEWED RATE
TRANSACTION ACTIVITY OVER TIME		TRANSACTION BY TYPE	
TRANSACTION BY CHANNELS		TRANSACTIONS STATUS	
CUSTOMER SEGMENT ANALYSIS		RISK REVIEW ANALYSIS	
DIGITAL ENGAGEMENT ANALYSIS			

6.6 Insights the Dashboard Is Expected to Communicate

The dashboard is intended to help users understand the overall level and value of transaction activity, how activity changes over time, which transaction types and channels are most commonly used, how transactions are performing, how behaviour differs between customer groups, and what patterns exist among transactions flagged for risk review.

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